IT BUSINESS SCHOOL · TUNISIA

Data Mining & Machine Learning · 2025-2026

US Housing Intelligence Platform

- Final report and Streamlit app defense
- Data Mining and Machine Learning project
- Ayoub Naouech and Rayen Belhadj

242
Observations

27
Features

20yr
Period

Ayoub Naouech · Rayen Belhadj

FINAL DEFENSE

Presentation Outline

01 **Project Motivation & Context**
Why housing markets matter

02 **Dataset & Features**
20 years, 242 observations, 27 variables

03 **CRISP-DM Methodology**
Full 6-phase data science workflow

04 **Streamlit Intelligence App**
Interactive dashboard screenshots

05 **Model Evaluation**
Metrics, diagnostics, and prediction proof

06 **OLAP & Paper Review**
Segmentation and theory validation

07 **Business Impact**
Stakeholders and decision support

08 **Limitations & Future Work**
Next development steps

09 **Final Conclusion**
Main defense message

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Project Context

- The U.S. housing market connects household wealth, credit conditions, inflation, construction, and employment.
- Housing prices react to demand, supply, income, mortgage rates, expectations, and timing.
- The project converts raw housing and macroeconomic data into an interactive intelligence platform.

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Problem Statement

- Main question: how can data mining and machine learning analyze and predict U.S. housing-market trends?
- Analytical challenge: housing variables interact in non-linear and time-dependent ways.
- Communication challenge: model results must be understandable for non-technical users.

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Project Objectives

- Analyze historical evolution of the Home Price Index.
- Identify variables associated with U.S. housing-price behavior.
- Train and compare machine learning models using reproducible metrics.
- Build a Streamlit app for visualization, modeling, OLAP, export, and interpretation.

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Dataset Overview

242

Monthly observations

27

Dataset columns

2004-2024

Historical period

- Target variable: Home_Price_Index.
- Main variables: mortgage rate, interest rate, unemployment, inflation, permits, income, population, and sentiment.
- The dataset supports both prediction and interpretation.

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Data Preparation

- Parse and sort the date column chronologically.
- Convert numeric columns into model-ready values.
- Check missing values and duplicate rows before modeling.
- Create administration labels, lag features, rolling features, ratios, and percentage changes.

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CRISP-DM Methodology

- Business understanding defines the housing problem.
- Data understanding checks quality and temporal structure.
- Data preparation builds usable analytical features.
- Modeling and evaluation compare algorithms.
- Deployment is delivered through Streamlit.

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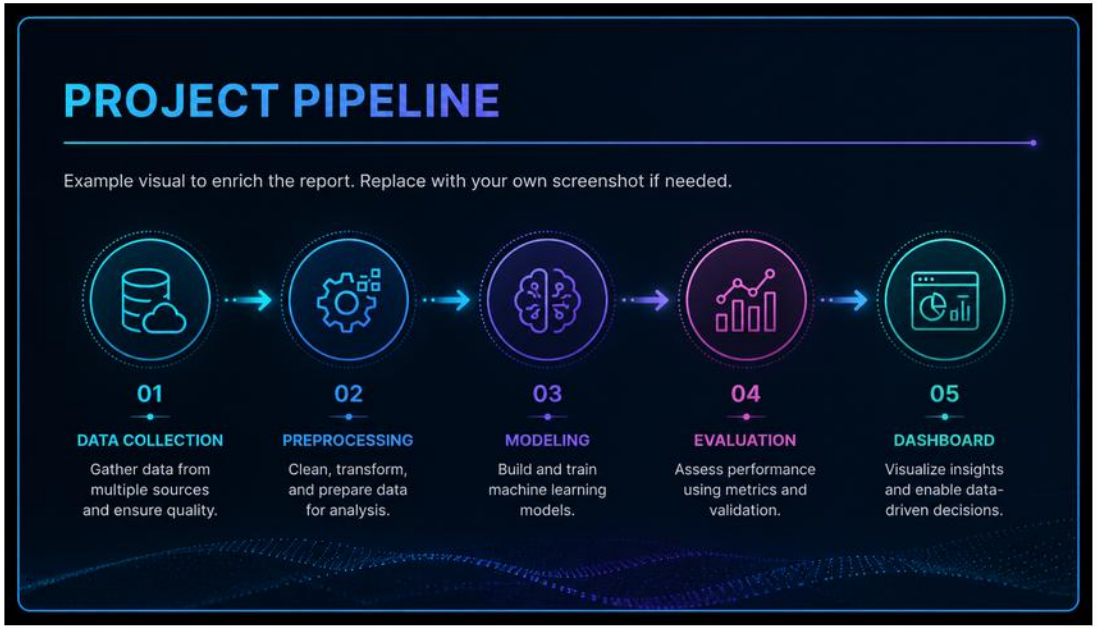
System Architecture



- Data layer: loading, cleaning, validation, and features.
- Analysis layer: EDA, correlations, OLAP, and period comparison.
- Modeling layer: regression, classification, forecasting, and diagnostics.
- Interpretation layer: paper review, conclusion, and export.

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Streamlit App: Home And Workflow

Ridge × RandomForest × LinearRegression × SVR × GradientBoosting ×

Run fit diagnostics

Diagnosis Numbers

Model	Task	Train score	Test score	Train-Test gap	Diagnosis	CV R2	CV std	Train MAE	Test MAE	Train RMSE	Test RMSE	What to fix
Ridge	Regression	0.9731	-1.8514	2.8246	Overfitting risk	-0.0277	0.7259	2.797	51.95	3.603	54.83	Increase `alpha` to add stronger regularization.
LinearRegression	Regression	0.9764	-3.2687	4.2451	Overfitting risk	0.0935	0.5843	2.635	63.106	3.377	67.086	Switch to Ridge/Lasso-style regularization or reduce high-le
GradientBoosting	Regression	0.9996	-4.2511	5.2507	Overfitting risk	-0.6148	1.988	0.309	67.408	0.42	74.406	Lower `learning_rate`, reduce `max_depth`, reduce `n_e
RandomForest	Regression	0.9925	-4.7585	5.751	Overfitting risk	-0.7183	2.3104	0.61	73.307	1.904	77.918	Reduce `max_depth`, increase `min_samples_leaf`, incre
SVR	Regression	0.8978	-10.386	11.2838	Overfitting risk	-3.2083	6.3021	4.574	103.949	7.026	109.565	Reduce `C`, increase `epsilon`, tune `gamma`, and scale

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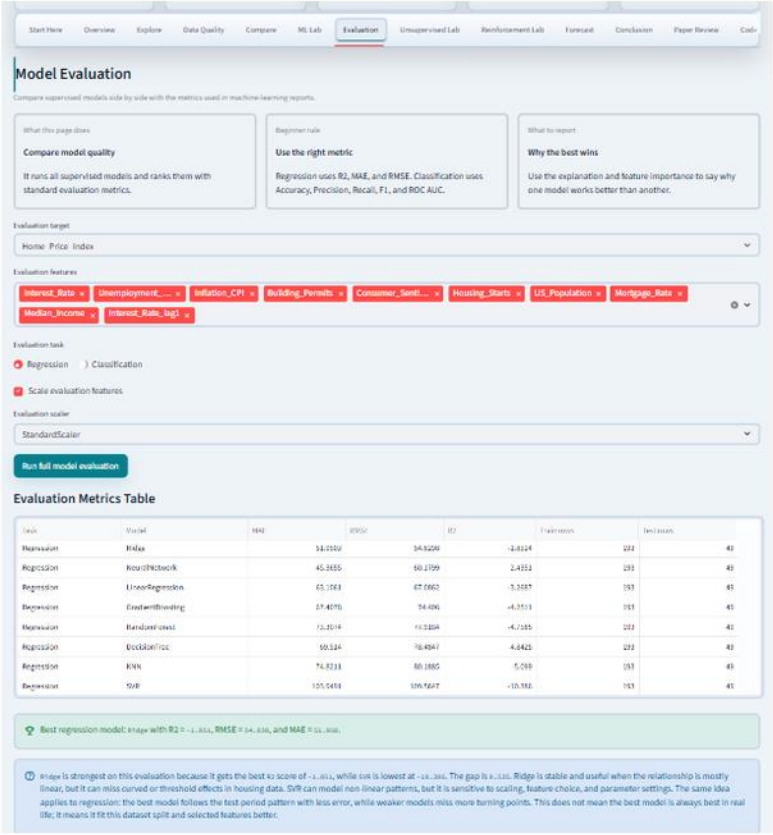
Streamlit App: ML Lab



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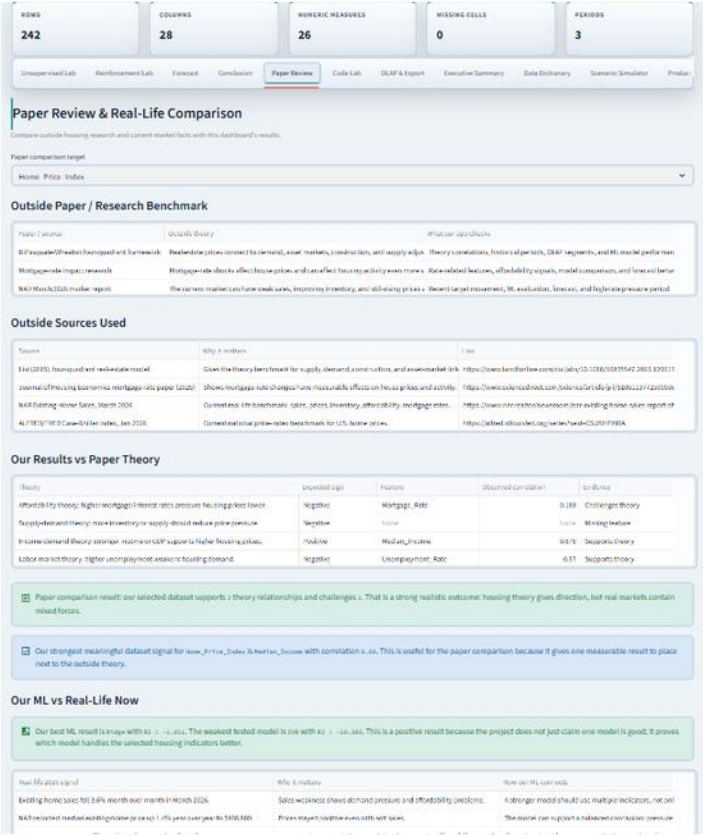
Streamlit App: Evaluation



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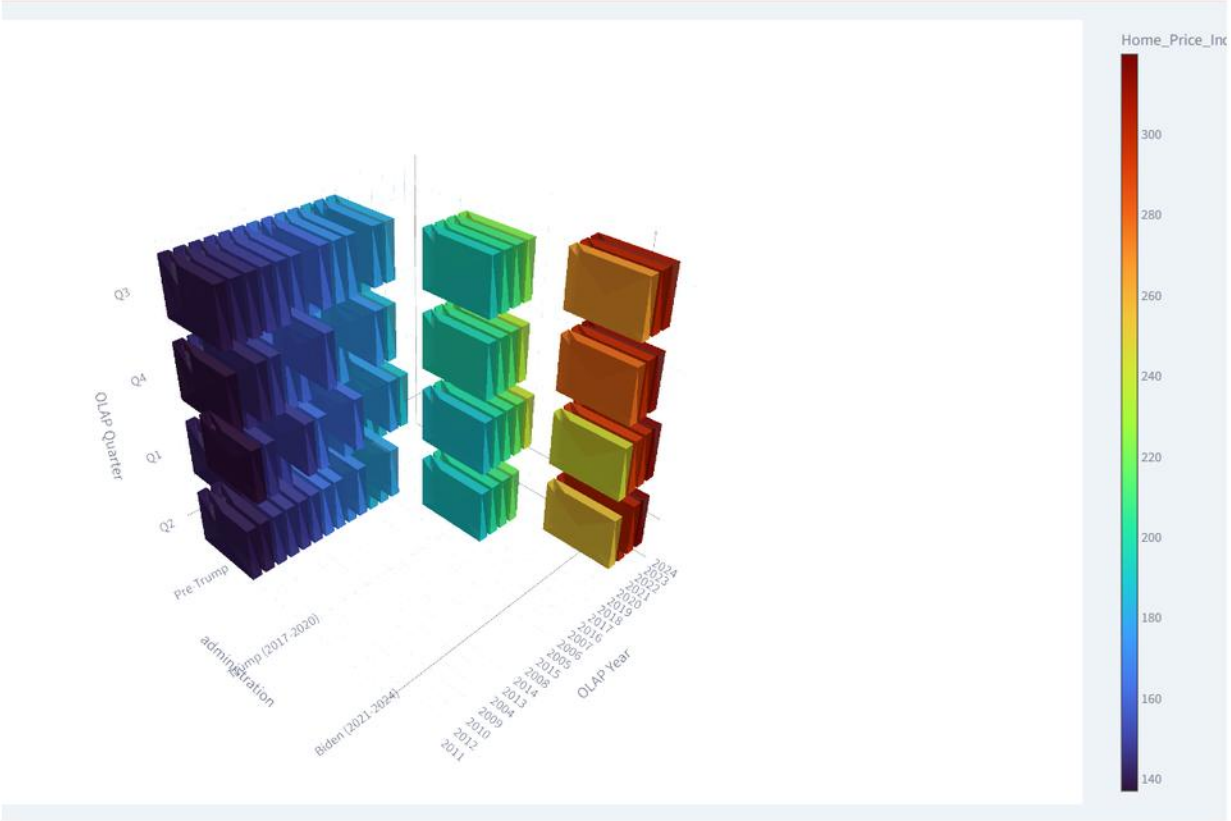
Streamlit App: OLAP And Export



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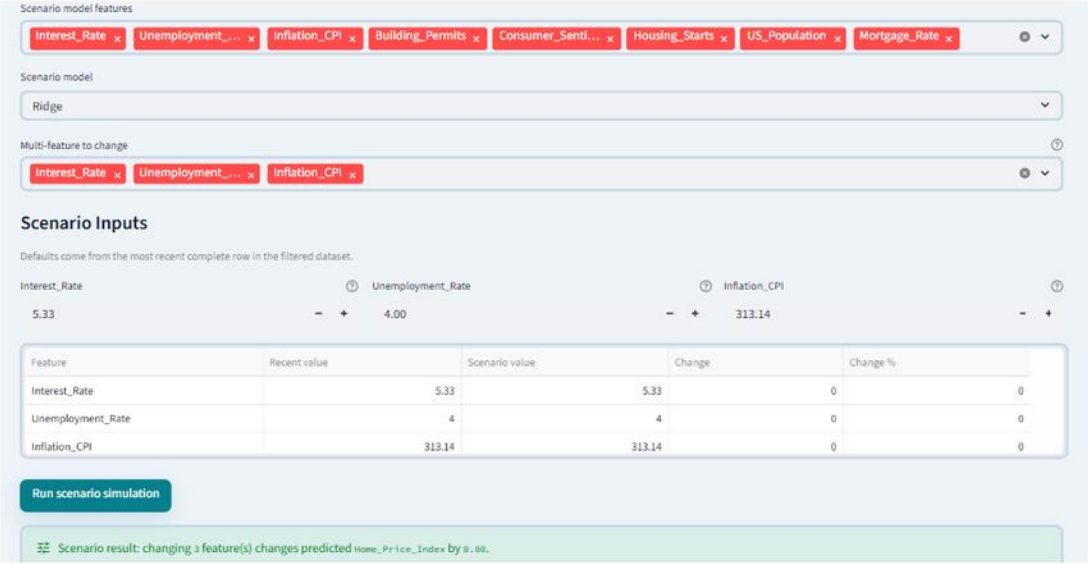
Streamlit App: Paper Review



- Compares dashboard evidence with housing theory.
- Explains rates, supply, demand, timing, and affordability together.
- Makes the project academically stronger than a purely technical dashboard.

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Streamlit App: Governance

Diagnosis Numbers

Model	Task	Train score	Test score	Train-Test gap	Diagnosis	CV R2	CV std	Train MAE	Test MAE	Train RMSE	Test RMSE	Hyperparameters	What to fix
Ridge	Regression	0.9731	-1.8514	2.8246	Overfitting risk	-0.0277	0.7259	2.797	51.95	3.603	54.83	Default	Increase `alpha` to add strong
LinearRegression	Regression	0.9764	-3.2687	4.2451	Overfitting risk	0.0935	0.5843	2.635	63.106	3.377	67.086	Default	Switch to Ridge/Lasso-style re
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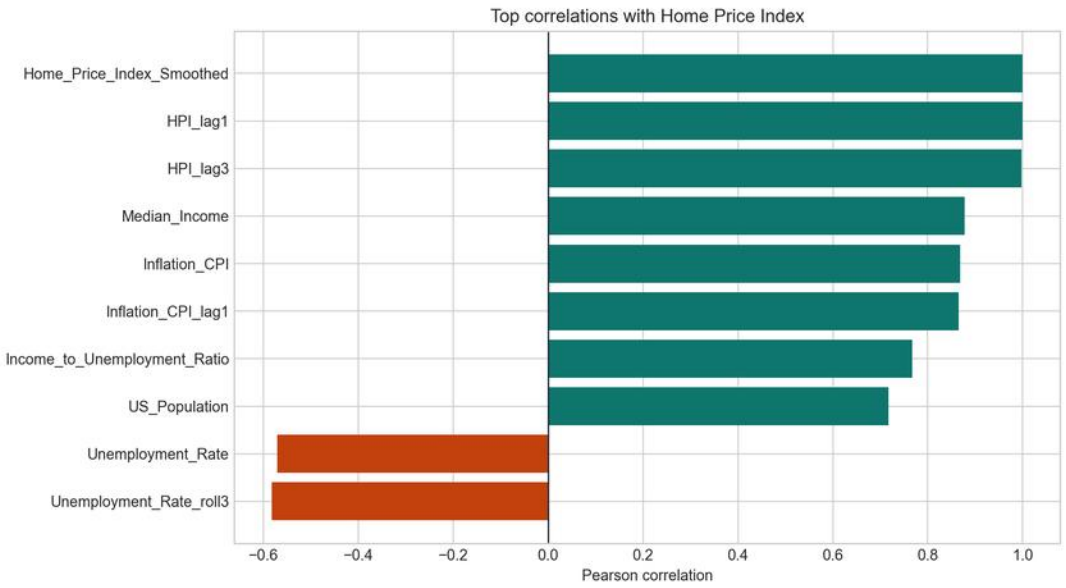
Correlation Results



- Strong relationships appear around lagged HPI, smoothed HPI, and Median_Income.
- Correlation supports interpretation but does not prove causality.
- Housing prices are persistent and linked to broader economic context.

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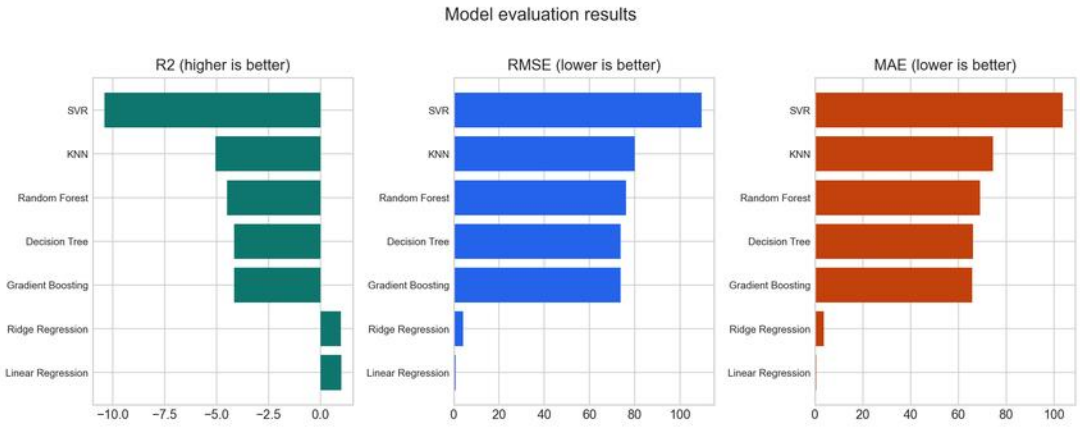
Model Evaluation Results



- Compared models include Linear Regression, Ridge, Random Forest, Gradient Boosting, Decision Tree, KNN, and SVR.
- Best generated metrics: R2 = 0.999, RMSE = 0.797, MAE = 0.661.

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Regression Metrics Summary

R2 0.999

Linear Regression

Best generated test result

R2 0.984

Ridge Regression

Stable interpretable baseline

Gap risk

Tree Models

Useful but weak extrapolation

Key message: model evaluation must be chronological and diagnostic, not based on popularity.

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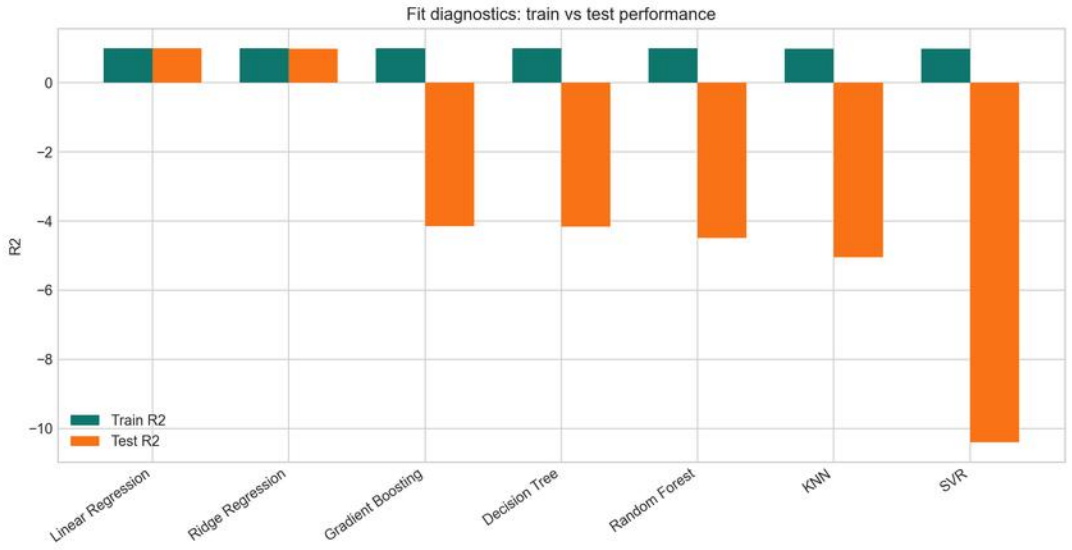
Fit Diagnostics



- Diagnostics compare train and test performance.
- Large train-test gaps indicate weak generalization or overfitting risk.
- Tree models struggle when the final period moves beyond the training range.

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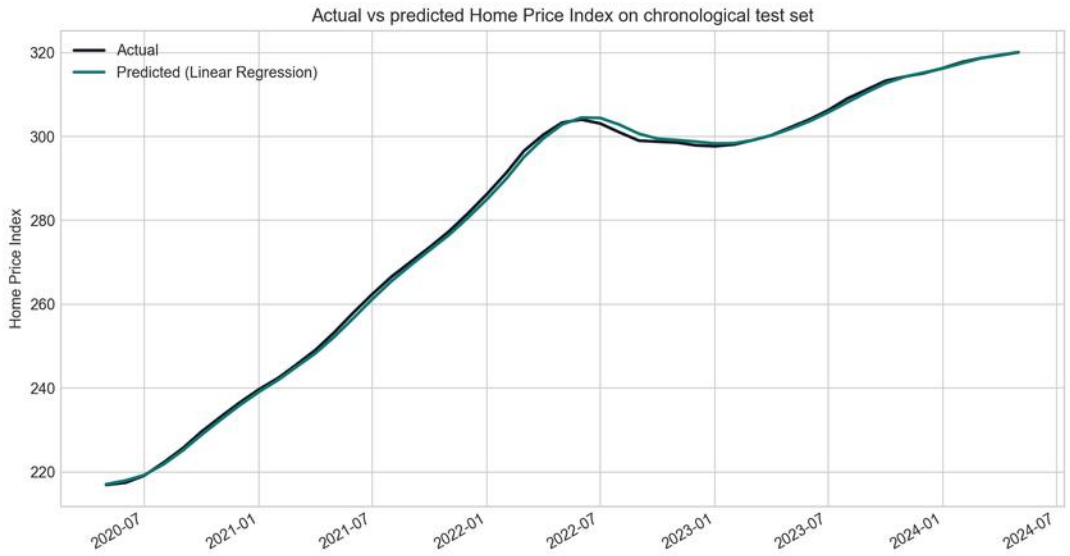
Actual Versus Predicted



- The best model follows the chronological test period closely.
- Lagged and smoothed price features explain the strong predictive result.
- The visual is easier to defend than metrics alone.

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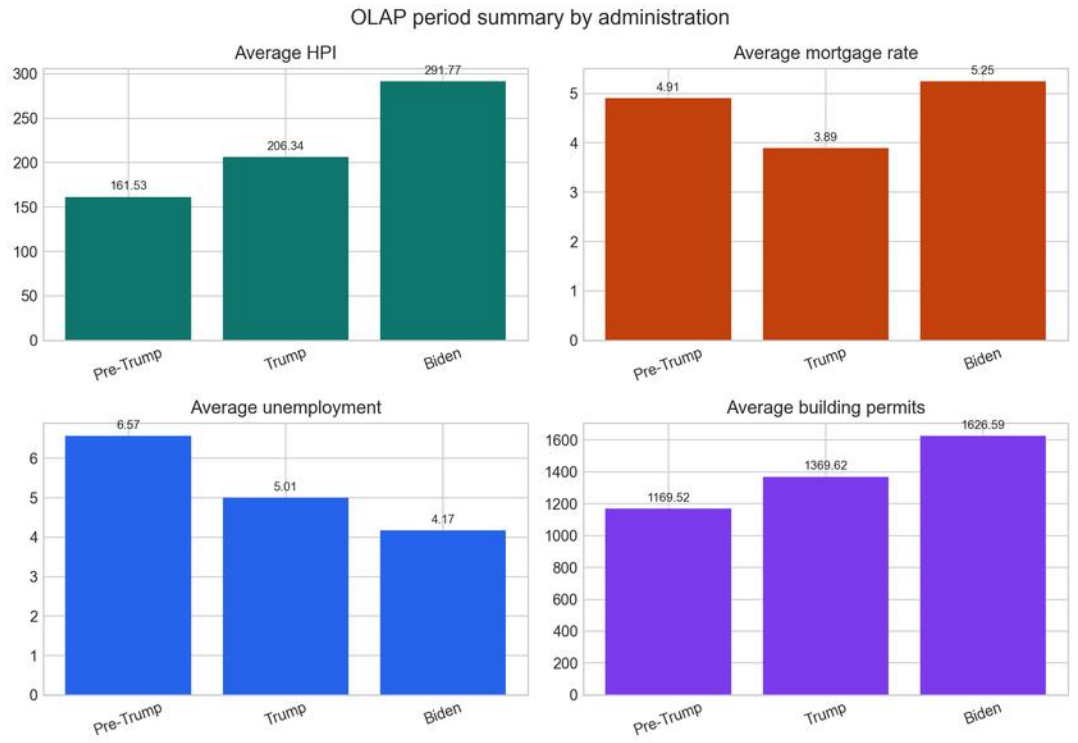
OLAP Period Results



- Biden period has the highest average Home Price Index.
- Biden period also has the highest average mortgage-rate pressure.
- Prices and affordability pressure can rise together when supply and timing matter.

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Paper Review Message

- Housing prices are shaped by demand, financing conditions, supply, construction, income, and timing.
- Higher mortgage rates reduce affordability, but prices may react slowly when supply is constrained.
- The strongest conclusion is multi-factor interpretation rather than one-variable explanation.

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Business Impact

- Analytical impact: users understand which indicators move with housing prices.
- Decision-support impact: users compare models, periods, OLAP segments, and scenarios.
- Educational impact: the project demonstrates EDA, ML, forecasting, OLAP, paper review, and governance.

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Limitations And Future Work

- National-level data hides regional differences between states and cities.
- Forecasts are educational and should not be treated as financial advice.
- Future work: regional datasets, live APIs, SHAP explainability, better chatbot grounding, monitoring, and authentication.

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Final Conclusion

- The report, poster, and app now tell one consistent story.
- U.S. housing prices require theory, correlations, model evaluation, diagnostics, OLAP, and market context.
- The Streamlit dashboard is a housing intelligence platform, not only a collection of charts.